

ANSHU GUPTA

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Summary

Ph.D. candidate in Computer Science and Engineering at UC San Diego specializing in **High-Performance Computing (HPC) and hardware-software co-design** for large-scale bioinformatics and AI workloads. Led **ROADIES** (PNAS, journal cover; 176x speedup) and **DP-HLS** (HPCA 2026; 32x speedup) from concept to publication. Seeking research scientist/engineer roles applying hardware acceleration to ML and computational biology.

Education

University of California San Diego **September 2022 – Present**
Ph.D. candidate in Computer Science and Engineering, (M.S. awarded, GPA: 4.0) La Jolla, CA

Indian Institute of Engineering Science and Technology, Shibpur **July 2015 – May 2019**
B.Tech. in Electronics and Telecommunication Engineering; GPA - 9.55/10 (Rank: 4) West Bengal, India
• DAAD WISE Scholar — summer research internship, University of Bremen, Germany

Relevant Publications [[Google Scholar](#)]

- **A. Gupta**, Y. Cao, J. Liang, Y. Turakhia, “DP-HLS: A High-Level Synthesis Framework for Accelerating Dynamic Programming Algorithms in Bioinformatics” - accepted at HPCA 2026. [[arXiv](#)]
- **A. Gupta**, S. Mirarab, Y. Turakhia, “Accurate, scalable, and fully automated inference of species trees from raw genome assemblies using ROADIES”, Proc. Natl. Acad. Sci. U.S.A. 122 (19), e2500553122 (2025). [[paper](#)]

Work Experience

AMD Research and Advanced Development (RAD) **June 2025 – Sep 2025**
Research Associate - PhD (Mentor: Stephen Neuendorffer) San Jose, California
• Emulated sub-modules of an advanced AI test chip on cutting-edge FPGA platform, contributing to the hardware and software bring-up of custom silicon.
• Conducted runtime performance characterization of AI models on the target test-chip architecture.

Analog Devices, Inc. **June 2023 – Sep 2023**
ASIC BU Digital Design Intern Wilmington, Massachusetts
• Conducted hardware architectural benchmarking for Cadence Tensilica Xtensa configurable processors.
• Characterized performance, power, and area metrics for low-power embedded processor configurations.

Analog Devices India Pvt. Ltd. **July 2019 – July 2022**
Digital Design Engineer Bengaluru, India
• Optimized embedded signal processing kernels using SIMD operations and architecture-specific vectorization, reducing DSP cycle count by 33%.
• Contributed to the digital front-end design and tape-out of a low-power Battery Management SoC.
• Evaluated ARM Cortex-M processor architecture and SRAM/ROM architecture’s power, performance, and area trade-offs for low-power embedded SoCs.

Research Experience (Mentor: Yatish Turakhia)

ML Hardware Acceleration **May 2025 – Ongoing**
• Designing **hardware-aware weight compression techniques** to significantly optimize and accelerate Deep Learning inference on specialized hardware.
• Research focused on maximizing computational efficiency and reducing memory footprint for deploying large-scale neural networks at the edge.

ROADIES (Automated Species Tree Inference Tool for Genomic Datasets) **Jan 2023 – Ongoing**

- **Led the development of ROADIES**, a reference-free, highly parallelized HPC tool for evolutionary genomics using **GPU/CPU co-processing and parallelization primitives**.
- Achieved **176x speedup** in tree generation for large-scale genomics dataset, reducing computational cost.
- Published in PNAS, featured on the [Journal Cover](#).
- Presented at **Intelligent Systems for Molecular Biology (ISMB) 2024** conference.

DP-HLS (FPGA Acceleration Framework)

Nov 2022 – June 2024

- **Developed DP-HLS**, an HLS-based framework, that achieved a **32x speedup** and **20x faster implementation time** for accelerating dynamic programming algorithms for bioinformatics workloads.
- Designed the framework for easy customization of algorithms without hardware redesign, using highly parameterizable systolic array architectures.

Technical Skills

GPU & HPC Acceleration: CUDA, MPI, OpenMP, Intel TBB, Slurm, Amazon AWS, Snakemake, Conda, Bash

AI/ML & Software: Python, C/C++, HLS, R

Hardware & Co-Design: ASIC/Custom Silicon Design Flow, FPGA (AMD/Xilinx Vitis HLS/Vivado), Verilog, SystemVerilog

Leadership & Service

- **Artifact Evaluation Committee:** ISCA 2025, ISCA 2024 | **Reviewer:** *Bioinformatics* (Oxford Academic) | **Workshop Co-organizer:** BioSys Workshop @ ASPLOS 2024, San Diego, CA

Other Projects

Parallelized Genomic and HPC Algorithms | C++, CUDA, Intel TBB, Git Jan 2023 – March 2024

- Executed high-performance parallelization of Suffix Array construction on GPUs, achieving 86 – 571x speedup over optimized CPU baselines for genomic read mapping.
- Accelerated dense matrix multiplication using CUDA on NVIDIA K80/T4 GPUs and implemented Intel AVX2 vectorization for high-throughput computing
- Optimized computation on the Expanse Supercomputer by applying MPI and C++ parallelization to challenging solvers (e.g., Aliev-Panfilov)

Optimization of HLS4ML Library | Vitis HLS, Vivado, Git, Python

April 2023 – June 2023

- Improved the HLS4ML library to achieve efficient ML hardware inference via High-Level Synthesis, focusing directly on the DL-to-FPGA flow.
- Partnered with CERN's HLS4ML team to implement feature enhancements and optimizations.

Relevant Coursework

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| • Principles of Computer Architecture | • Design Automation and Prototyping |
| • Parallel Computer Architecture | • Validation and Testing in Embedded Systems |
| • Parallel Computation | • Algorithms in Computational Biology |
| • Parallel Computation in Bioinformatics | • Computational Evolutionary Biology |